

Terraform Practical

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1. State files in Terraform

- Using the script below, Terraform will create an EC2 instance named **MyFirstInstance** with an identity of **id01** and instance type **t3.micro**.

```
root@ip-172-31-8-242:/home/ubuntu/terraform_directory# ls
aws  awsclicv2.zip  configuration.tf  ec2.tf  terraform.tfstate  terraform.tfstate.backup
root@ip-172-31-8-242:/home/ubuntu/terraform_directory# cat ec2.tf
resource "aws_instance" "id01" {
  ami           = "ami-073130f74f5ffb161"
  instance_type = "t3.micro"
  subnet_id    = "subnet-0cbc6ee27b50c1eda"
  vpc_security_group_ids = ["sg-06c3ae8be5e2b7113"]

  tags = {
    Name = "MyFirstInstance"
  }
}
```

Instances (1) [Info](#)

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☐	Name 	Instance ID	Instance state	Instance type	Status check
☐	MyFirstInstance	i-08350cd91dbff973c	 Running  	t3.micro	 3/3 checks passed

- If the instance is modified outside of Terraform for example, changing the instance type and running *terraform apply* again will reconcile the actual state with the desired configuration, ensuring the infrastructure matches the definitions in the HCL script.

Change instance type [Info](#) | [Get advice](#)

You can change the instance type only if the current instance type and the instance type tha

Instance ID

 [i-08350cd91dbff973c](#) (MyFirstInstance)

Current instance type

t3.micro

New instance type

✕

Change instance type to: t3.micro

c7i-flex.large

m7i-flex.large

t3.micro

t3.small

5 (32-bit) / (64-bit) architecture can be chosen.

this instance type

t3.micro

☒	Name 	Instance ID	Instance state 	Instance type 	Status check
☒	MyFirstInstance	i-08350cd91dbff973c	 Running  	t3.small	 Initializing

Plan: 0 to add, 1 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

```
aws_instance.id01: Modifying... [id=i-08350cd91dbff973c]
aws_instance.id01: Still modifying... [id=i-08350cd91dbff973c, 00m10s elapsed]
aws_instance.id01: Still modifying... [id=i-08350cd91dbff973c, 00m20s elapsed]
aws_instance.id01: Still modifying... [id=i-08350cd91dbff973c, 00m30s elapsed]
aws_instance.id01: Still modifying... [id=i-08350cd91dbff973c, 00m40s elapsed]
aws_instance.id01: Still modifying... [id=i-08350cd91dbff973c, 00m50s elapsed]
aws_instance.id01: Still modifying... [id=i-08350cd91dbff973c, 01m00s elapsed]
aws_instance.id01: Still modifying... [id=i-08350cd91dbff973c, 01m10s elapsed]
aws_instance.id01: Modifications complete after 1m12s [id=i-08350cd91dbff973c]
```

Apply complete! Resources: 0 added, 1 changed, 0 destroyed.

root@ip-172-31-8-242:/home/ubuntu/terraform directory#

☒	Name 	Instance ID	Instance state 	Instance type 	Status check
☒	MyFirstInstance	i-08350cd91dbff973c	 Running  	t3.micro	 Initializing

- Terraform maintains a **state file**, named **terraform.tfstate** by default, which is stored in the same directory where Terraform is executed. This file is automatically created after running *terraform apply* and contains a JSON-formatted mapping of the resources defined in the configuration to the real resources in your infrastructure. It is crucial for tracking and managing infrastructure changes.